ML Report

Ayaulym Raikhankyzy

May 2026

1 Agentic RAG Implementation

Standard RAG follows a fixed retrieve → augment → generate chain. Agentic RAG instead lets the LLM control retrieval: it rewrites the query, retrieves evidence, judges quality, and retries when evidence is weak [5, 4]. This is useful for ICD-10 coding because patient complaints are written in common Russian, while protocols use technical clinical Russian.

We compare five configurations: an *LLM-only* baseline, a *Simple* retrieval baseline, and three full agentic systems: `rag_complex`, `rag_advanced`, and `rag_hybrid`. The three full systems share the same LangGraph agent shell and differ only in the `retrieve` node.

1.1 Shared Agent Architecture

The full agent uses seven nodes: *rewrite*, *HyDE*, *retrieve*, *grade*, *rewrite_retry*, *candidates*, and *select*. Gemma-4-26B-A4B-it performs query rewriting, HyDE generation [3], retrieval grading [5], and final code selection using structured chain-of-thought reasoning [4].

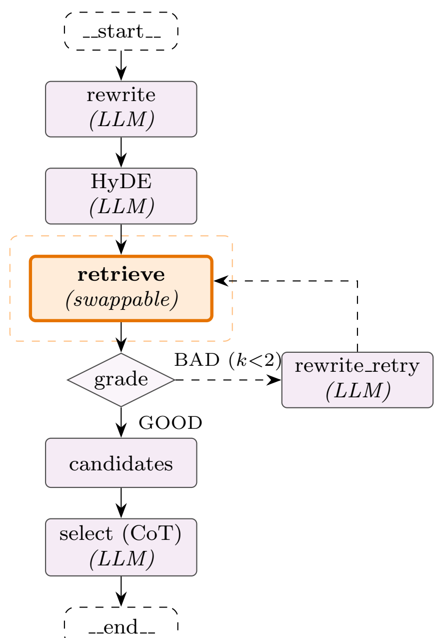

Figure 1: Shared agentic architecture used by `rag_complex`, `rag_advanced` and `rag_hybrid`. Solid arrows are the nominal path; dashed arrows are the CRAG self-correction loop (up to two retries). Only the dashed orange `retrieve` block differs across pipelines. The *Simple* baseline removes the *HyDE*, *grade*, and *rewrite_retry* nodes; the *LLM-only* baseline removes everything except the *select* node.

The flow is: rewrite the complaint, generate a HyDE protocol excerpt, retrieve top-5 protocols, grade evidence, retry if needed, aggregate up to 40 ICD-10 candidates, and select one final code.

1.2 Baselines and Retrieval Variants

LLM-only sends the complaint directly to the LLM and asks for one ICD-10 code. **Simple** uses rewrite → retrieve → candidates → select with the same retriever as `rag_complex`, but without HyDE or CRAG retry.

Table 1: Retrieval modules used in the three full agentic pipelines.

Pipeline	Retriever	Ranking	ICD Aggregation
`rag_complex`	BGE-M3 dense + BM25 sparse	RRF over dense query, dense HyDE, and BM25; BGE reranker	Codes from top-5 protocols using metadata and harvested descriptions
`rag_advanced`	BGE-M3 chunk-level dense search with FAISS `IndexFlatIP`	BGE cross-encoder on retrieved chunks	Codes in chunk text get full score; metadata-only codes get 0.05× score
`rag_hybrid`	Fine-tuned multilingual-e5-base + ICD graph	FAISS dense search; no reranker	$0.7s_{\text{semantic}} + 0.3s_{\text{graph}}$ plus keyword boosts

In short, `rag_complex` emphasizes strong fusion and reranking, `rag_advanced` emphasizes chunk-level code-aware aggregation, and `rag_hybrid` tests whether an ICD graph can compensate for a smaller retriever.

1.3 Results

Table 2 reports results on the 221-case Russian benchmark. Baselines were evaluated only on headline accuracy metrics.

Table 2: Results. Best values are bolded.

Pipeline	S@1	L@1	R@3	MRR	NDCG	P@5	C-Rec
LLM-only	13.64	39.09	13.63	–	–	–	–
Simple	23.53	43.44	28.41	–	–	–	–
`rag_hybrid`	23.53	48.42	28.05	0.2826	0.3054	39.82	51.13
`rag_advanced`	28.96	**52.94**	**35.75**	**0.3513**	**0.3802**	52.49	**61.54**
`rag_complex`	**29.86**	51.13	34.84	0.3473	0.3725	**54.30**	57.92

S@1 = Strict@1; L@1 = Lenient@1; R@3 = Recall@3; P@5 = Protocol Recall@5; C-Rec = Candidate Recall.

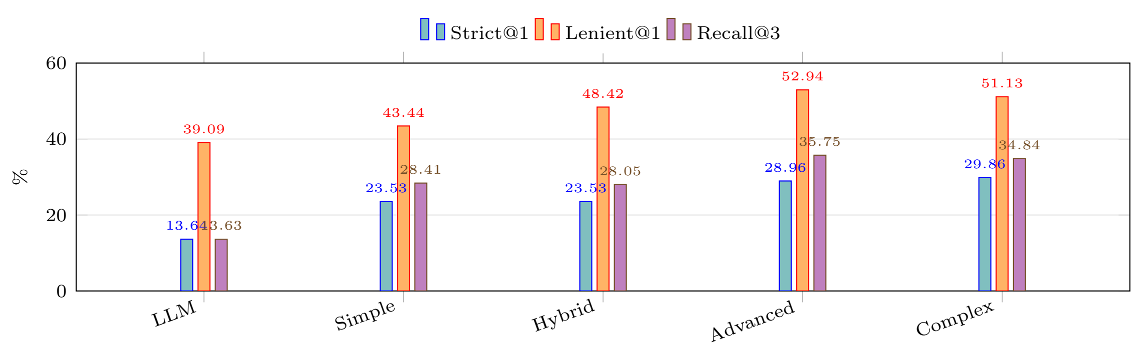

Figure 2: Headline accuracy across all five configurations.

1.4 Discussion

Retrieval gives the largest gain. LLM-only reaches only 13.64% Strict@1 and 13.63% Recall@3. Adding retrieval through the Simple baseline raises Strict@1 to 23.53% and Recall@3 to 28.41%. This shows that grounding in clinical protocols is more important than relying on the LLM's parametric knowledge.

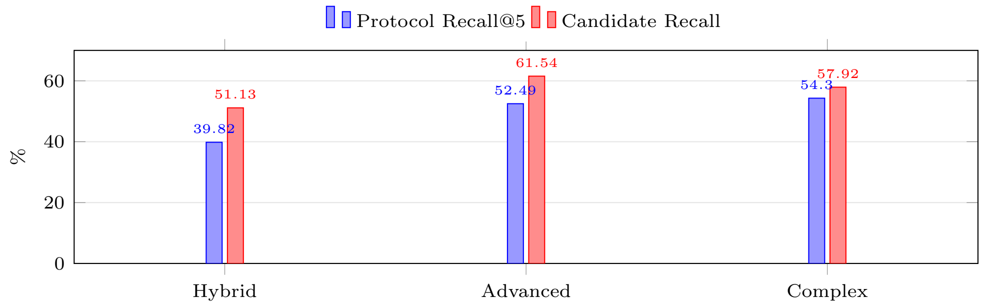

Figure 3: Retrieval quality for the three full agentic pipelines.

HyDE and CRAG give a smaller but useful gain. Compared with Simple, `rag_complex` adds HyDE and the grader/retry loop. This improves Strict@1 by +6.33 pp, Lenient@1 by +7.69 pp, and Recall@3 by +6.43 pp. Thus, agentic self-correction helps, but its effect is smaller than the initial gain from retrieval.

Advanced retrieves better; Complex selects better. `rag_complex` achieves the best Strict@1 and Protocol Recall@5, suggesting stronger top-rank selection. `rag_advanced` achieves the best Lenient@1, Recall@3, MRR, NDCG@10, and Candidate Recall, suggesting better candidate generation. In short: `rag_advanced` retrieves better, while `rag_complex` selects better.

Hybrid is limited by the smaller retriever. `rag_hybrid` uses multilingual-e5-base instead of BGE-M3 and does not include a cross-encoder reranker. Its ICD graph helps, but not enough to overcome the weaker retriever, consistent with graph-RAG findings in medical retrieval [1].

The main bottleneck is selection. Candidate Recall is much higher than Strict@1 across all full pipelines. For example, `rag_advanced` includes the correct code in the candidate set for 61.54% of cases, but selects it first only 28.96% of the time. This suggests that a stronger selector or learned candidate reranker may improve performance more than further retrieval tuning.

Lenient accuracy may be near a ceiling. Lenient@1 stays within 48–53% for the three full pipelines. Breaking this band likely requires fine-tuning the LLM on ICD reasoning or adding clearer family-level diagnostic distinctions to the corpus.

References

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[5] J. Gu et al., "A survey on LLM-as-a-judge," *The Innovation*, p. 101253, 2026.